Switching Gene Expression On and Off

Biology 2e chapter 16 reading connection | Teacher guide / classroom preparation

Learning target

How can cells carry a reporter gene without visibly expressing it?

Timing and materials

Setup class and observations after incubation, commonly on a following day.

Educational arabinose-controlled fluorescent reporter kit or teacher-prepared approved reporter cultures, appropriate agar plates and controls, sterile loops and pipettes, incubator or kit-approved room-temperature schedule, compatible viewer, approved culture disposal supplies.

Classroom route

1. Use only a school-approved educational strain and kit protocol. Teacher prepares cultures/plates and follows the supplier's incubation and disposal requirements. If transforming cells, use the official transformation protocol and include its controls.

2. Compare no-plasmid cells on nonselective and antibiotic plates with reporter-bearing cells on antibiotic plates with and without arabinose. Match inoculum and handling; replicate plates when supplies allow.

3. Observe growth and fluorescence separately after the specified incubation. Keep plates closed during student observation; use the viewer's protective shield. Document confluent growth rather than inventing a colony count.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

The no-plasmid nutrient plate supports growth; the no-plasmid antibiotic plate does not in this model. Reporter-bearing cells grow on both selective media but fluoresce with arabinose. Growth and induction are separate observations. The model starts with prepared cells, so it does not estimate transformation efficiency.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Classification feedback is a practice aid and does not overwrite student scores.

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Model limitations

Generic arabinose-regulated reporter, not the lac operon. Colony appearance is a simplified visibility model, not bacterial generation time. Supplied prepared cells omit transformation; a single plate per condition cannot estimate variability.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Counting activities require every specimen plus a review; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Optional pGLO protocol and equipment

https://commerce.bio-rad.com/webroot/web/pdf/lse/literature/1660033_EDU.pdf

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Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.